

ORIGINAL ARTICLE

Influence of Coverage and Diversity of Submerged Macrophytes on the Sediment Bacterial Community in Dongshan Bay, Lake Taihu

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ABSTRACT

1. Submerged macrophytes influence sediment microbial communities in shallow lakes; however, the ecological roles of their community-level characteristics remain underexplored. In this study, we examined the relationships between macrophyte community coverage and diversity and the composition of sediment bacterial communities in Dongshan Bay, Lake Taihu.
2. We combined high-throughput sequencing, functional prediction, and multivariate statistical modelling to analyse sediment bacterial composition, potential metabolic functions, community assembly processes, and their associations with macrophyte communities and environmental factors.
3. Macrophyte community structure explained more variation in sediment bacterial community structure than seasonal dynamics. Higher coverage and diversity were related to improved water clarity, elevated sediment nutrient concentrations, and enhanced bacterial metabolic potential. In macrophyte-dominated zones, bacterial communities were primarily structured by homogeneous selection, and microbial co-occurrence networks became less complex.
4. Our results indicate that species richness, diversity, and coverage most effectively capture the ecological functions of submerged macrophyte communities, and collectively influence sediment bacterial communities through multiple pathways, including habitat modification and altered selection pressures.
5. This study underscores the ecological role of macrophyte diversity and coverage in shaping sediment microbiota, offering insights for lake restoration strategies.

1 | Introduction

Submerged macrophyte communities are key components of lake ecosystems (Andersen et al. 2020; Dong et al. 2014; Wang et al. 2008). They directly absorb nutrients and stabilise the sediments, thus helping to mitigate water eutrophication (Gao et al. 2023; Wang et al. 2008). However, negative effects might arise if plant biomass is not harvested (Gong ChongXian and Wang Dong 2014; Shi et al. 2022; Song et al. 2013). Dead plant biomass accumulates on the sediment surface, releasing nutrients upon decomposition (Eid et al. 2014; Lan et al. 2012). This excessive sedimentary organic carbon enrichment depletes oxygen, degrades benthic habitat quality, and might increase methane emissions (Gremmen et al. 2023; Grutters et al. 2017). These multiple effects of submerged macrophytes on the environment are largely determined by the properties of plant communities. For example, high-density and high-coverage plant communities can more comprehensively cover the surface sediment, reducing water turbulence (Gao et al. 2020). Studies also indicate that plant diversity contributes most to water quality due to their multifaceted impacts on ecological functions (Li et al. 2023; Lv et al. 2019). Plants with strong and extensive stem systems (such as *Potamogeton maackianus*) are more effective in stabilising sediments compared to those with fragile or sparse stem systems (Bertrin et al. 2017; Schallenberg and Sorrell 2009). Additionally, some submerged plants enhance phosphorus co-precipitation from the water column by altering water alkalinity through photosynthesis (Jerauld et al. 2024; Kufel and Rymuza 2014).

Submerged macrophytes influence sediment bacterial communities through multiple pathways (Chao et al. 2021; Gao et al. 2023). Studies have indicated that submerged plants are highly selective of their associated epiphytes and rhizosphere microorganisms (Fan et al. 2016; Liu, Guan, et al. 2018; Zhang et al. 2018; Zhang, Zhang, et al. 2024). For example, in a mesocosm experiment, the relative abundance of Anaerolineae was significantly increased by the presence of *Ceratophyllum demersum*, while *Vallisneria spiralis* promoted the growth of Deltaproteobacteria and Gammaproteobacteria (Dai et al. 2019). However, most existing studies focus on the control of eutrophication through the restoration of specific submerged plants or the contribution of certain submerged plant species to certain element cycles (Liu et al. 2022; Xu et al. 2021). As submerged plants usually grow in communities in natural waters, it is of great importance to study the effect of submerged plant community properties (e.g., Shannon diversity and coverage) on sediment microbial community structure and assembly.

The ecological processes that shape sediment microbial communities in submerged macrophyte-dominated ecosystems remain largely undefined. Null model-based community assembly analysis is used to determine whether observed ecological patterns in community composition differ significantly from what would be expected by random chance (Shi et al. 2016; Song et al. 2023; Stegen, Freestone, et al. 2013; Stegen, Lin, et al. 2013). Investigating microbial co-occurrence patterns provides valuable insights into ecosystem functioning, revealing interactions among microbial species, such as cooperation or competition, which might influence nutrient cycling and organic matter decomposition (Faust and Raes 2012; Faust et al. 2012). Additionally, it helps to elucidate microbial network

dynamics, identifying keystone species that maintain community stability and offering predictions on how microbial communities may respond to environmental changes or disturbances (Chaffron et al. 2010). These insights are crucial for managing and restoring ecosystems effectively (Wang et al. 2024; Zhang, Zhou, et al. 2024).

Dongshan Bay is located in the eastern part of Lake Taihu and is highly colonised by native submerged macrophytes. Therefore, it is an ideal area to explore the feedback effects of long-established submerged macrophyte communities on environmental parameters and sediment bacterial communities. Elucidating the interplay among sedimentary microbial co-occurrence patterns, community assembly mechanisms, and functional potential in relation to submerged macrophyte community assemblages and environmental variables in Dongshan Bay could yield critical insights into the long-term ecological implications of aquatic vegetation dynamics on lacustrine biogeochemical cycling processes. In this study, we acquired the biomass, coverage, and diversity indices of submerged macrophyte communities as well as the water and sediment parameters and sediment bacterial community information from 15 sample plots in Dongshan Bay in spring (germination period), summer (growing period), and winter (apoptosis period), to:

1. Elucidate the impacts of submerged macrophytes on water and sediment properties in Dongshan Bay.
2. Identify indicators (e.g., coverage, biomass, diversity) that reflect the influence of submerged macrophyte communities on the surrounding environmental conditions, including water quality, sediment characteristics, and microbial composition.
3. Elucidate how macrophyte communities shift the sediment microbial community through direct and indirect effects in Dongshan Bay.

Our results provide insights into the long-term impacts of submerged macrophyte restoration on and offer theoretical support for large lakes' ecological remediation.

2 | Materials and Methods

2.1 | Study Area and Sample Collection

Lake Taihu is the third largest freshwater lake in China, located in the southern part of the Yangtze River Delta, with a surface area of 2338 km² and an average water depth of 2.3 m (Song et al. 2009). Historically, Lake Taihu had abundant aquatic plants, but these have receded since the 1970s due to the construction of flood dikes. Our sampling work was conducted in January (winter), April (spring), and June (summer) of 2021 in Dongshan Bay. A total of 15 sample plots were established. Each sample plot covered an area of 16 m² and contained four sites (Figure 1). The water depth (m) was measured in situ at each sample site using a plastic rod with a scale and averaged for each sample plot. Integrated water samples were collected using a 5 L plastic sampler in the middle of each sample plot. In each of the sample plots, surface sediment samples (about 0–1 cm depth) were collected from four

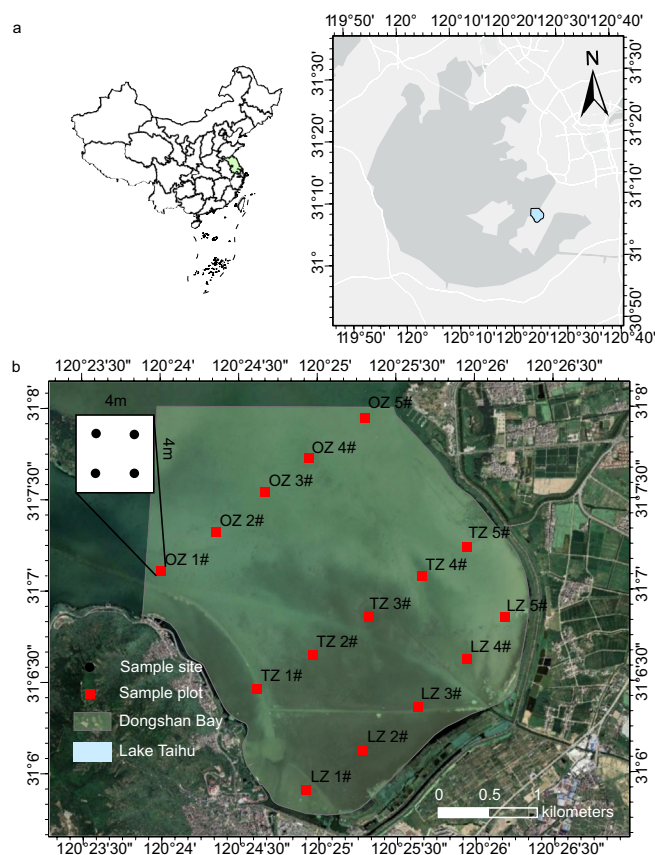


FIGURE 1 | (a) Each sample plot with the area of 16 m² and contain 4 sample sites. (b) the distribution of sample plots in Dongshan Bay. (c) the Location of Dongshan Bay and Lake Taihu.

evenly distributed sites (replicates) using a custom-designed grab sampler (Figure S2). The four sediment subsamples were then thoroughly homogenised to generate a single composite sample per plot. This approach was used to reduce micro-scale heterogeneity and to represent the overall sediment condition at the sample plot level, which served as the primary spatial unit for microbial community analysis. The four sediment samples were then thoroughly homogenised to generate a single composite sample per plot. This approach was used to reduce micro-scale heterogeneity and to represent the overall sediment condition at the sample plot level, which served as the primary spatial unit for microbial community analysis. Part of the fresh sediments was used for physicochemical parameter measurements, and the remaining sediments were freeze-dried and stored at -80°C for high-throughput sequencing of 16S rRNA genes. In total, 45 water, sediment, submerged macrophyte, and sedimentary microbial samples were obtained.

Lake Taihu experiences dominant southeast winds from spring to summer and northwest winds from autumn to winter. Wind-induced currents play a significant role in driving turbulence within the water body (Wang et al. 2016), causing sediment resuspension and movement toward downwind areas, which results in uneven underwater topography across the bay (Figure S1). Additionally, we observed that submerged plants tend to grow in the shallow zones along the windward shore (Figure S1), likely due to the shallow water column and higher light availability

(Dong et al. 2021; Liu, Yao, et al. 2018). Thus, we grouped the sample sites into shallow zone (SZ), central zone (CZ), and bay mouth zone (MZ) according to the distance from the windward shore (D, m), water depth gradient, and submerged macrophyte distribution of each plot, with five plots arranged in each zone (Figure 1).

2.2 | Submerged Macrophyte, Water and Sediment Parameters Determination

The plant coverage (coverage-p) in each plot was estimated using the Braun-Blanquet method (Pirini et al. 2011; Ziaja et al. 2021). Submerged macrophyte and surface sediment samples were quantitatively collected using a clip-type sampler (CN111044314A, Figure S2), a custom-designed sampling device for collecting macrophytes and sediments. The number of species, abundance (plant/m²) and biomass (biomass-p, kg/m²) of plant samples were determined immediately. The plant number data was used to calculate the diversity indices, including species richness (species number-p), Simpson evenness (Simpson-p), and Shannon diversity (Shannon-p). The plant parameters from the four sample sites within the same plot were averaged for further analysis.

Water samples were stored in a refrigerator prior to analysis to minimise microbial activity and chemical alteration. Unfiltered water was used for the analysis of total nitrogen (TN-w, mg/L) and total phosphorus (TP-w, mg/L), while filtered samples (0.45 μm membrane) were used for the determination of nitrate (NO_3^- -w, mg/L), nitrite (NO_2^- -w, $\mu\text{g/L}$), ammonia (NH_4^+ -w, mg/L), and orthophosphate (SRP-w, $\mu\text{g/L}$), following standard methods (Gilcreas 1966). The chlorophyll *a* (Chl.a-w, mg/L) concentration was calculated from spectrophotometric measurements after extraction in 90% hot ethanol (Chen et al. 2003). Here, the suffix '-w' denotes concentrations in the water column.

The fresh sediment was centrifuged at 4000 rpm for 10 min to obtain interstitial water, which was passed through a 0.45 μm pore filter membrane. The obtained filtrate was used for the determination of sediment ammonia (NH_4^+ -s, mg/kg), sediment nitrate (NO_3^- -s, mg/kg), sediment nitrite (NO_2^- -s, mg/kg), and sediment total nitrogen (TN-s, mg/kg). The total organic carbon concentration in sediment (TOC-s, mg/kg) was determined using the potassium dichromate redox method. Sediment phosphorus forms were classified into loosely sorbed phosphorus (L-P-s, mg/kg), iron-bound phosphorus (Fe-P-s), aluminium-bound phosphorus (Al-P-s, mg/kg), organic phosphorus (O-P-s, mg/kg), calcium-bound phosphorus (Ca-P-s, mg/kg), and residual phosphorus (Res-P-s, mg/kg) based on the hierarchical extraction method of (Rydin 2000). Here, the suffix '-s' denotes concentrations in sediment pore water.

2.3 | DNA Extraction, PCR, and Illumina Sequencing

Total genomic DNA was extracted from sediment samples using the PowerSoil DNA Isolation Kit (MoBio Laboratories, USA) according to the manufacturer's instructions. All DNA samples

were stored at -80°C until further analysis. The V4 region of the bacterial 16S rRNA gene was amplified using the universal primers 515F (5'-GTGCCAGCMGCCGCTAA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3'). PCR was performed in 20 μL reaction volumes containing 5 $\times$ FastPfu Buffer, dNTPs, primers, FastPfu Polymerase, BSA, and template DNA. The thermal cycling conditions were as follows: 95°C for 3 min; 27 cycles of 95°C for 30 s, 55°C for 30 s, 72°C for 30 s; followed by a final extension at 72°C for 10 min.

PCR products were pooled and purified using the AxyPrep DNA Gel Extraction Kit, and sequencing was carried out on the Illumina MiSeq platform. Raw reads were processed using the QIIME 2 pipeline. Sequences were quality-filtered, clustered into Operational Taxonomic Units (OTUs) at 97% sequence similarity, and taxonomically assigned using the SILVA database (version 138). OTUs, which represent groups of closely related sequences as proxies for microbial taxa, were used for downstream community analysis. Taxa belonging to Bacteria and Archaea were filtered using the 'dplyr' package in R. The raw sequence data have been deposited in the NCBI Sequence Read Archive (SRA) under accession number PRJNA1161583.

2.4 | Analysis of Microbial Community Composition and Function

Microbial community composition and diversity were assessed based on the OTU data generated from QIIME 2. Alpha diversity indices, including the number of OTUs and the Shannon index, were calculated to evaluate within-sample diversity. Beta diversity patterns were examined using non-metric multidimensional scaling (NMDS) based on Bray–Curtis dissimilarities. To assess differences in community composition among the shallow zone (SZ), central zone (CZ), and bay mouth zone (MZ), *t*-tests were performed on the sample scores along the NMDS axes. Both NMDS and *t*-tests were conducted using the 'vegan' package in R. To predict the functional composition of microbial communities, the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway profiles were inferred using PICRUSt2 (<http://picrust.github.io/picrust/>), based on 16S rRNA gene data.

2.5 | Identifying Variables Associated With Sediment Bacterial Communities and Their Assembly Processes

We conducted Partial Least Squares Path Modelling (PLS-PM) to identify variables associated with sedimentary bacterial communities and to identify the relationships between these variables and sediment microbial communities in Dongshan Bay using the 'plspm' package in R software. We hypothesized that the macrophyte community could influence water and sediment parameters, subsequently affecting the sediment bacterial community. Direct and indirect dependencies between response variables (the principal components of microbial community, NMDS 1 and NMDS 2) and all measured parameters were assessed in PLS-PM using path analysis. To simplify the model, we clustered observed variables into five

latent variables: submerged macrophyte community, hydrodynamic conditions which include water depth and D as key indicators of flow intensity and spatial variation (Kalfi 2002; Ramón et al. 2022), water parameters, sediment parameters, and the β -diversity of sediment bacterial community (represented by NMDS 1 and NMDS 2). Initial models were constructed using all samples and included all plausible pathways between variables (Figure S3). Then, we optimized the fitting effect by adjusting the path. The final model was selected based on the highest goodness of fit index (GoF). The variables in the final model were selected as the variables most strongly influencing bacterial communities, and the importance of each observed variable on latent variables was estimated by their weight values. Relationships between latent variables were estimated via path effects. The relationships between latent variables and non-affiliated observed variables were estimated via cross-loading value. Furthermore, we used these variables in principal component analysis (PCA) to indicate the gradient of environmental changes related to submerged macrophyte communities. The PCA was conducted using 'vegan' package in R software.

To define potential community assembly processes, the 'picante' R package (version 4.1.2) was used to implement the null model. βNTI (Beta Nearest Taxon Index) was calculated to assess the contribution of deterministic vs. random processes in community assembly. The model classified community assembly into five major ecological processes: heterogeneous selection ($\beta\text{NTI} < 2$), homogeneous selection ($\beta\text{NTI} > 2$), homogeneous dispersal ($|\beta\text{NTI}| < 2$ and $\text{RCbray} < -0.95$), dispersal limitation ($|\beta\text{NTI}| < 2$ and $\text{RCbray} > -0.95$), and undominated fraction ($|\beta\text{NTI}| < 2$ and $|\text{RCbray}| < -0.95$) (Stegen, Lin, et al. 2013).

2.6 | Network Construction

Network analysis has proven helpful in deciphering complex microbial interaction patterns (Faust and Raes 2012; Zhang, Chen, et al. 2021). We constructed an integrated network based on all microbial samples, extracting the network's topological parameters to describe microbial community complexity. Microbial networks were built using the 'igraph' and 'psych' packages. To reduce noise and false-positive predictions, we restricted our analysis to taxa with abundance $> 0.05\%$. Pairwise Spearman correlations between OTUs were calculated, and *p*-values were adjusted using the Benjamini and Hochberg false discovery rate (FDR) test. Only robust correlations ($|r| > 0.60$) and statistically significant (FDR-adjusted *p*-value < 0.01) correlations were retained in network analyses (JagadeeswaraRao and SivaPrasad 2024).

The network topological parameters of each sample (sub-network) were extracted using the subgraph function in the 'igraph' package and the methods of (Zhang et al. 2018). The topological parameters of the integrated network and sub-networks used in this study included node number, average connectivity, degree of centralization, clustering coefficient, centralization of degree, density, and average path length. Additionally, two key node characteristics derived from network modularity were within-module connectivity (Z_i) and

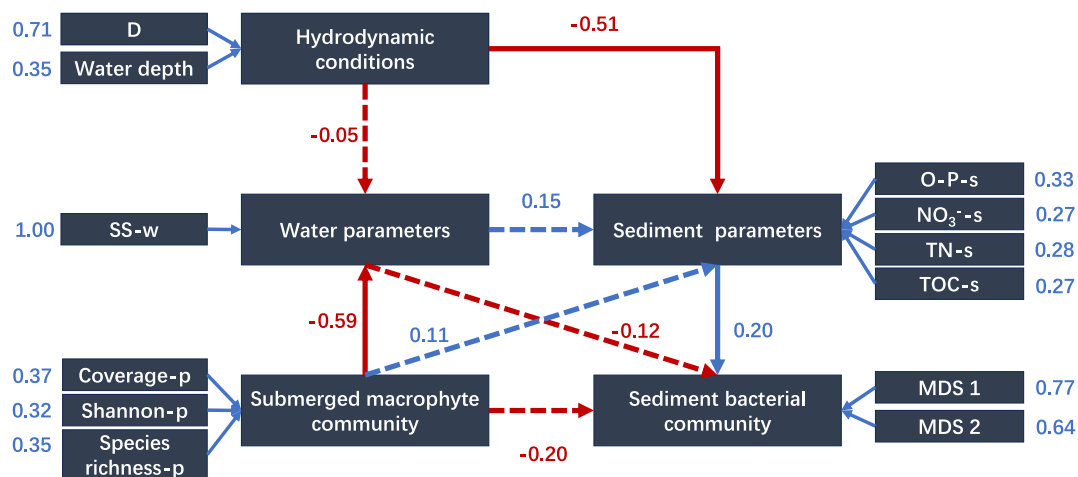


FIGURE 2 | Pathway Relationships between environmental variables, submerged macrophyte community properties and sediment bacterial community. The solid lines represent significant pathway, the dash lines represent insignificant path way. The red and blue line represent positive and negative relationships, respectively.

between-module connectivity (Pi). Keystone species within the network were identified based on Zi and Pi values using the 'igraph' R package.

3 | Results

3.1 | Submerged Macrophytes Affect the Water Quality, Sediment Composition, and Structure of Bacterial Communities

A total of 12 submerged macrophyte species were detected in Dongshan Bay: *Potamogeton wrightii*, *Potamogeton maackianus*, *Stuckenia pectinata*, *Myriophyllum verticillatum*, *Myriophyllum spicatum*, *Vallisneria natans*, *Cabomba caroliniana*, *Najas minor*, *Chara vaillant*, *Ceratophyllum demersum*, *Hydrilla verticillata*, and *Elodea canadensis*. Among these, *Potamogeton wrightii*, *Hydrilla verticillata*, and *Vallisneria natans* were the most commonly found species in sample plots. Submerged plants exhibited diverse distribution and varying coverage in Dongshan Bay. We observed that the coverage, biomass, Shannon index, reversed Simpson index, and species number of submerged macrophyte communities decreased from SZ to MZ (Table S1), indicating that the shallow zone is more suitable for the growth of submerged aquatic plants. Simultaneously, we observed decreasing trends of total nitrogen (TN-w), soluble reactive phosphorus (SRP-w), and suspended solids (SS-w) in water from the MZ to the SZ. In contrast, all sedimentary nutrient concentrations were significantly higher in the SZ compared to the CZ and MZ.

We hypothesized that submerged macrophyte communities strongly influence the environment, thereby affecting microbial communities. A PLS-PM approach was developed to select variables most associated with sedimentary bacterial communities and elucidate relationships between sediment bacterial communities, environmental factors, and submerged macrophyte community structures. The initial model (Figure S3) included all variables and had a GoF of 0.32. After changing the pathways and variables, the final model, with a GoF of 0.51 (Figure 2), was

obtained. The final model identified important environmental variables and revealed the causal relationships between latent variables. Hydrodynamic conditions were determined by D value and water depth (weights=0.71 and 0.35, respectively). Coverage-p, species richness-p, and Shannon-p were identified as the key indicators of submerged macrophyte-related ecological effects (weights=0.37, 0.35, and 0.32, respectively). Hydrodynamic conditions and submerged macrophyte communities collectively influenced water parameters (determined by SS-w, with path effects = -0.05 and -0.59, respectively) and sediment parameters (path effects = -0.51 and 0.11, respectively). The sedimentary bacterial community was primarily associated with submerged macrophyte communities (path effect = -0.21), followed by sediment parameters and water parameters (path effects = 0.20 and -0.12, respectively).

3.2 | The Relationship Between Sediment Bacterial Communities, Environmental Factors, and Submerged Macrophyte Communities

A total of 25,296 OTUs were detected in Dongshan Bay. The OTU richness in SZ was significantly lower than in CZ ($p < 0.01$, Figure S4). The bacterial Shannon index in the MZ and CZ was significantly higher than those in the SZ (Figure S4, $p < 0.01$). NMDS and *T*-test results indicated that sediment bacterial community compositions in the three areas were significantly separated along NMDS 1 ($p < 0.01$, Figure 3). Notably, the separation between zones (e.g., SZ to MZ) was much more pronounced than the seasonal-based variation ($p > 0.05$, Figure S5).

To further characterise the dominant taxa across zones, Proteobacteria (relative abundances of 28.91%, 25.80%, and 29.00% in SZ, CZ, and MZ, respectively) and Acidobacteriota (9.73%, 11.20%, and 10.7% in SZ, CZ, and MZ, respectively) were the most abundant phyla (Figure S6), followed by Chloroflexi (10.32%, 10.30%, and 6.63% in SZ, CZ, and MZ, respectively) and Desulfobacterota (9.00%, 8.74%, and 7.41% in SZ, CZ, and MZ, respectively).

Based on the final PLS-PM, 10 variables were identified associated with microbial communities (Figure 2). Among them, organic phosphorus (O-P-s) and total nitrogen (TN-s) had the highest relationships on sediment bacterial communities (cross loadings=0.25 and 0.14, respectively), followed by the *D* value (cross loading=0.11). Submerged macrophyte community properties, including Coverage-p, Shannon-p, and species richness-p, also had considerable relationships on the bacterial communities (cross loadings=-0.08, -0.10, and -0.11, respectively). PCA and correlation analysis revealed potential correlations and coordinated changes among selected variables, with the first component (environment PC1) explaining 52.2% of the variation (Figure S7a) and showing significant differences between three areas (Figure S7c). Submerged macrophyte properties and sedimentary parameters were positively correlated with environment PC1, while water depth, *D* value, and suspended solids in water (SS-w) were tightly correlated with the negative ordination of environment PC1 (Figure S7b). There were significant differences in the projection of SZ, CZ, and MZ plots on PC1 (Figure 4 and Figure S7c), indicating distinct environmental gradients associated with

submerged macrophyte habitats and sediment microbial communities in Dongshan Bay. Based on these differences, we use the environmental PC1 axis to represent the gradient of submerged macrophyte-related environmental factors in the study area.

3.3 | Assembly Processes and Co-Occurrence Patterns of Sedimentary Bacterial Communities

The null model was used to determine the assembly mechanisms of sediment bacterial communities in Dongshan Bay (Figure 5a), distinguishing between selection and neutral processes by calculating standardised phylogenetic turnover (β NTI). The contribution of selection processes (heterogeneous and homogeneous selection) on community structure increased from MZ to SZ, while the contribution of neutral processes (dispersal limitation, homogenising dispersal, and others) showed an opposite trend. This suggests that environmental homogeneity and the similarity of sediment microbial communities decreased from SZ to MZ.

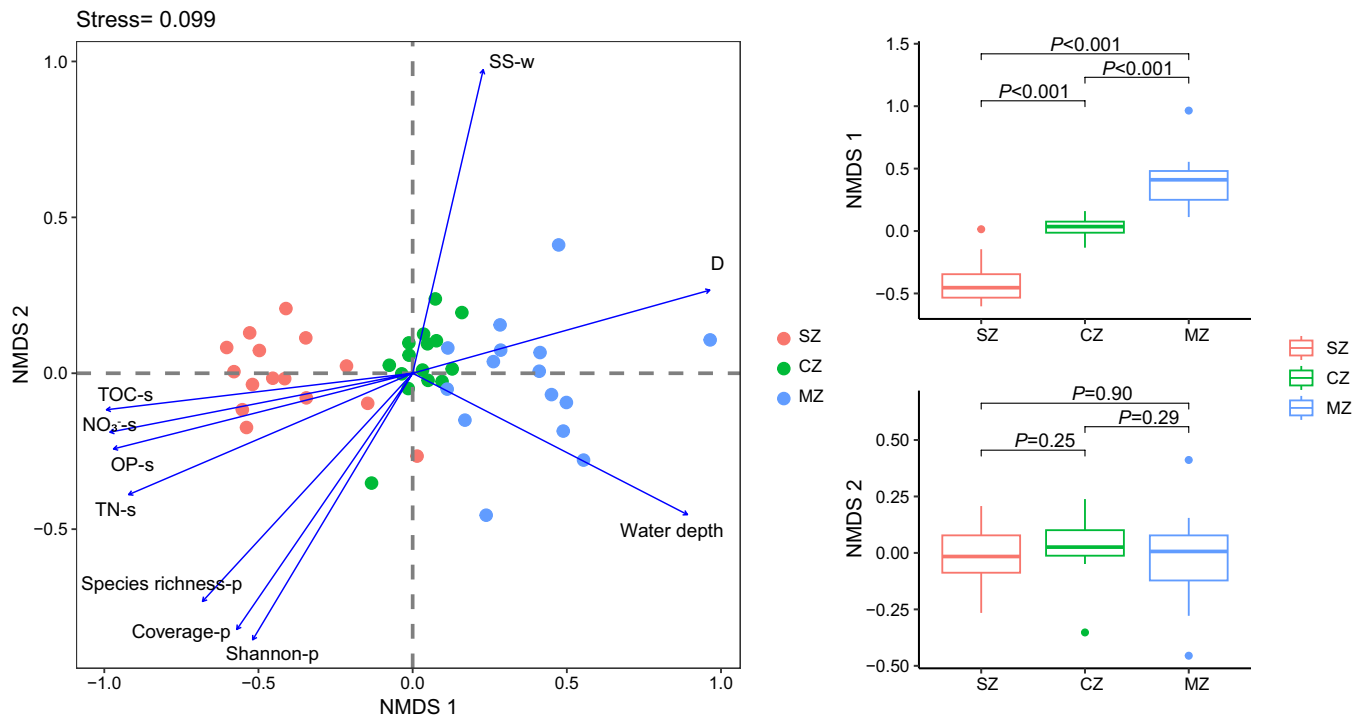


FIGURE 3 | NMDS and *T*-test shows the sediment bacterial communities in Dongshan Bay are well separated by LZ, TZ and OZ along NMDS1. The selected drivers are fitted in coordinate and are represented by blue solid line with arrow.

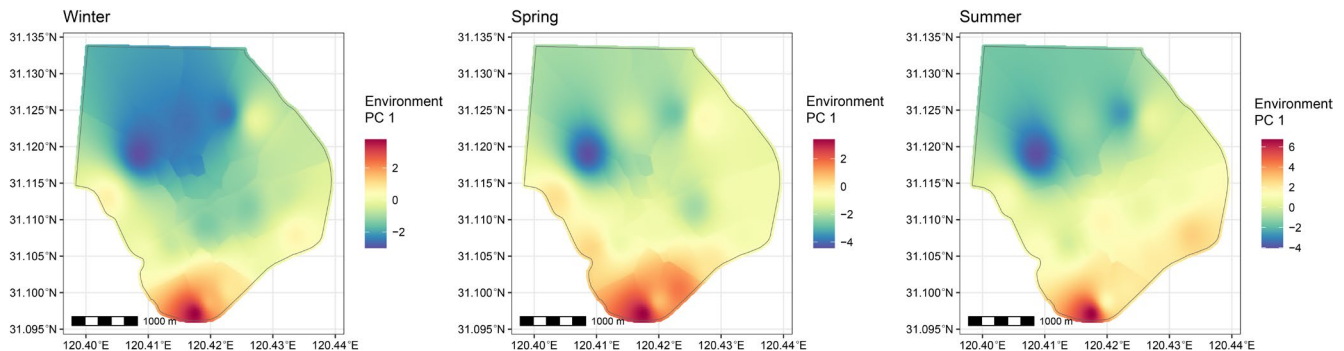


FIGURE 4 | Spatial variations of environmental PC1 in Dongshan Bay.

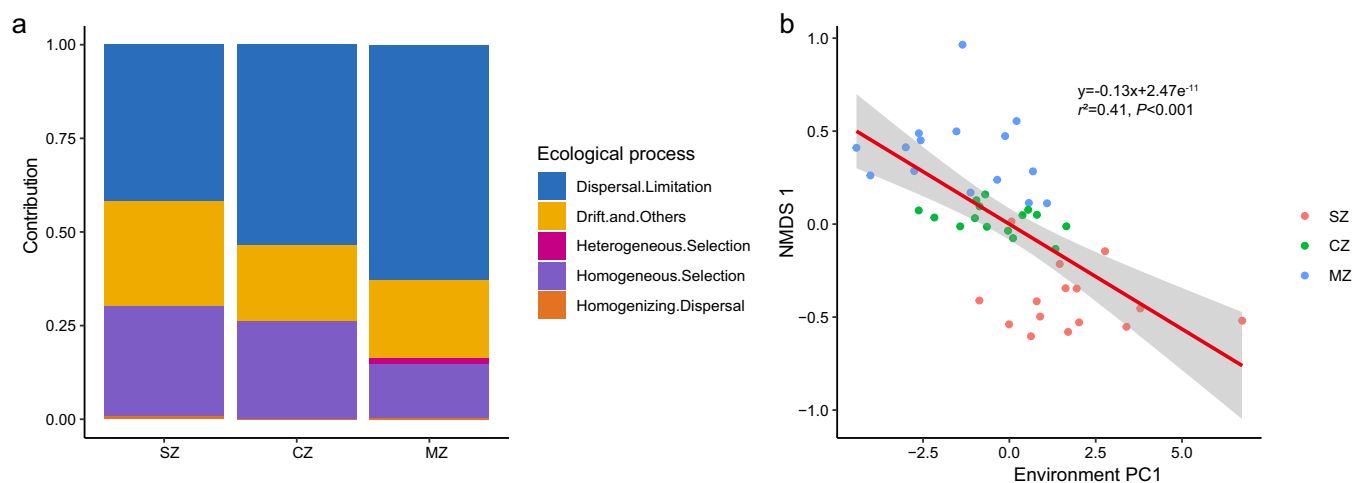


FIGURE 5 | (a) Ecological processes contributing to sediment microbial community assembly across different zones in Dongshan Bay. (b) Relationship between NMDS1 and environment PC1 in different zones in Dongshan Bay.

The integrated network captured 280,018 associations among 1440 bacterial OTUs, with degrees distributed according to a power-law distribution (Figure S8), indicating a scale-free distribution and non-random co-occurrence pattern. The sediment community network was clearly divided into five modules: taxa preferring high nutrient levels in Modules 1 and 2 (M1, M2), taxa preferring medium nutrient levels in Module 3, and taxa preferring low nutrient levels in Modules 4 and 5 (Figure 6a). Furthermore, Acidobacteriota and Proteobacteria had the top three highest degrees and node numbers in all modules. In contrast, Bacteroidota had the top three highest degrees and node numbers in M1 and M2, while Chloroflexi had the top three highest degrees and node numbers in M4 and M5 (Table S2). This indicates that Bacteroidota play an important role in maintaining taxa coexistence under high nutrient levels. In contrast, Chloroflexi are more important for maintaining coexistence under medium and low nutrient levels, as nodes with a higher degree of centrality are more crucial in sustaining taxa interactions within networks.

We next generated sub-networks for each sample by retaining OTUs associated with specific samples and all edges among them in the integrated network, thereby capturing different aspects of network topological features. Various topological features were calculated for these sub-networks, including node number, degree, average path length, betweenness centralization, network density, and clustering coefficient. We then applied a linear regression model to determine the relationships between co-occurrence patterns and environmental gradient indicators (environment PC1, Figure 6b).

Further, four taxa of keystone species were identified via Zi-Pi value (Figure S9a). Two of them belonged to Desulfobacterota (OTU_1255 and OTU_1700), and the rest belonged to Bacteroidota (OTU_16949) and Sva0485 (OTU_95). Keystone taxa are not highly abundant species within communities, but their essential ecological roles cannot be overlooked. We found the abundance of OTU_16949 and OTU_95 was significantly negatively correlated with environment PC 1 (Figure S9b).

3.4 | Prediction of Functional Dynamics of Sedimentary Bacterial Communities

Based on PICRUSt2 analysis and KEGG pathway annotation at level 1, five main functional categories were identified: brite hierarchies, cellular processes, environmental information processing, genetic information processing, and metabolism. Metabolism refers to microbial processes involved in energy production and nutrient cycling. Brite hierarchies include unclassified or broadly related metabolic functions. Cellular processes encompass fundamental cell functions such as cell division and signal transduction. Environmental information processing pertains to how microorganisms sense and respond to environmental changes. The most abundant functions were metabolism and brite hierarchies, accounting for approximately 36% and 34% of the total pathways, respectively. We used linear regression to examine the variation of KEGG pathways along the environmental gradient (environmental PC1); we found a significant increase in the pathways for carbohydrate metabolism, cell growth, and energy metabolism along environmental PC1 ($p < 0.05$, Figure 7). However, methane-related metabolic pathways, indicative of greenhouse gas emissions, showed no increase along environmental PC1 (Figure 7).

4 | Discussion

4.1 | Influence of Submerged Macrophytes on Water and Sediment Properties

Our study highlights the impacts of long-established submerged macrophytes on the physicochemical properties of water and sediments in Dongshan Bay. In eutrophic shallow lakes lacking submerged plants, the water in shallow zones typically exhibits higher turbidity and nutrient concentrations than the bay mouth zone, primarily due to cyanobacterial scum and sediment resuspension (Wang et al. 2016). However, the observed decreasing gradient in total nitrogen (TN-w), soluble reactive phosphorus (SRP-w), and suspended solids (SS-w) from the MZ to SZ underscores the crucial role of submerged macrophytes in nutrient uptake and sediment

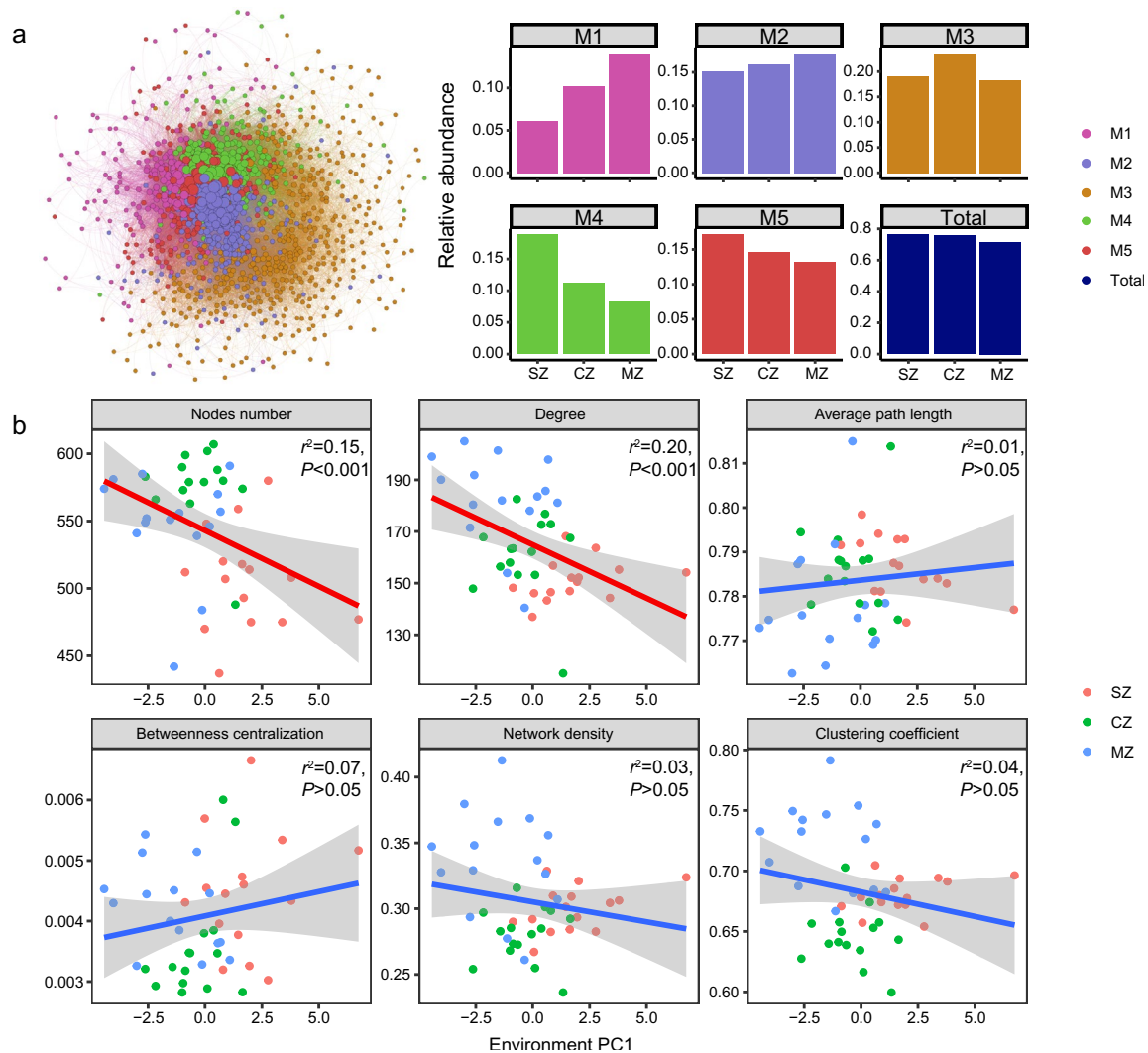


FIGURE 6 | Network Analysis of sediment microbial communities in Dongshan Bay: (a) Co-occurrence network modules and their relative abundance across different zones. (b) relationships between sub-network topological features and environmental PC1, determined by the regression coefficient of linear model.

stabilisation in Dongshan Bay. The results indicated that macrophyte coverage had the greatest contribution to the reduction of SS-w among the submerged macrophyte parameters (Pearson's $r^2 = -0.58$). Increasing the coverage of submerged plants dramatically reduces water disturbance and enhances water clarity (Lv et al. 2023; Madsen et al. 2001; Zhang, Zhang, et al. 2021). The dense plant structure (e.g., stems and roots) slows down water currents and wave action, promoting the settling of suspended particles and preventing sediment resuspension (Bertrin et al. 2017; Schallenberg and Sorrell 2009). In addition, submerged macrophytes serve as host substrates for epiphytic biofilms. These biofilms can act as an additional filter for nutrients and particles in the water column, thereby enhancing the water purification function of macrophyte communities.

Conversely, the nitrogen and phosphorus content in sediments significantly increased from MZ to SZ, with increasing plant diversity and coverage. This can be attributed to the dual processes of enhanced input of organic material, including settled suspended solids and the macrophyte detritus (Chen et al. 2022;

Liu et al. 2024). The results showed that two diversity indices were highly positively correlated with plant coverage ($r^2 = 0.85$, $p < 0.01$ and $r^2 = 0.86$, $p < 0.01$) and also had considerable contributions to nutrients. This suggests that higher plant diversity promotes greater resource use efficiency, leading to increased coverage and productivity (Gao et al. 2020; Huston 2014; Wang et al. 2022), which in turn enhances the capacity for trapping suspended particles (Zhang et al. 2019) and increases detritus input into the sediments.

4.2 | Sediment Microbial Community Dynamics

In our study, the variability in sediment microbial communities appears to be influenced more by the submerged macrophyte community, rather than by seasonal shifts. This may be attributed to the stronger selective pressures exerted by well-established macrophytes on sediment bacterial communities. In fact, sediment bacterial communities in shallow lakes were considerably more stable across seasons than planktonic communities, likely due to the buffering capacity of sediments and

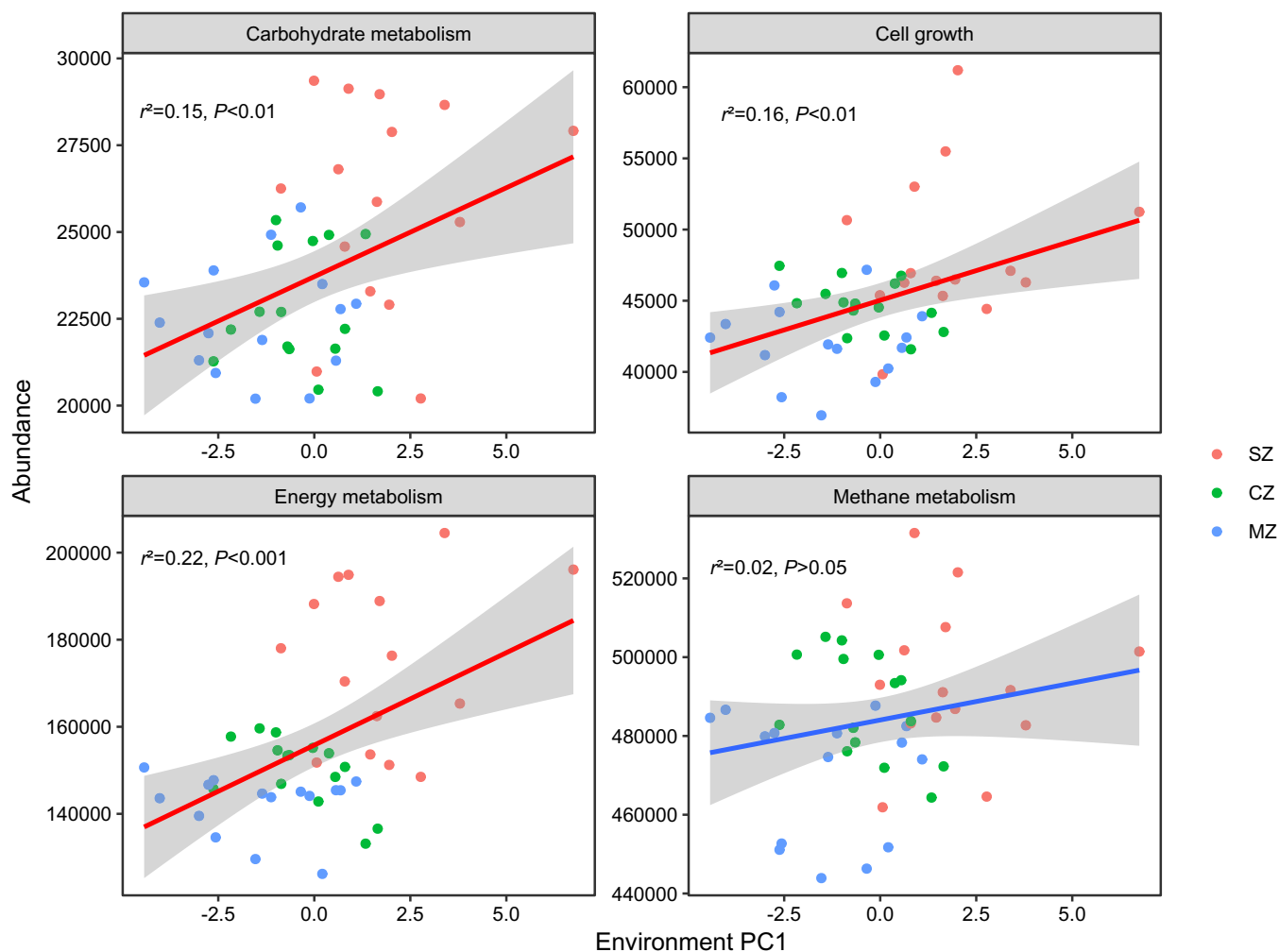


FIGURE 7 | Relationships between relative abundance of metabolic pathways and environmental PC1 in Dongshan Bay, determined by the regression coefficient of linear model.

the strong selection imposed by benthic environmental conditions (Jiao et al. 2021).

Key variables relating to microbial community structure include organic phosphorus (O-P-s), total nitrogen (TN-s), nitrate (NO_3^- -s), total organic carbon (TOC-s), macrophyte coverage and diversity, and hydrodynamic properties. These findings support the view that macrophytes create distinct habitats; therefore, shaping microbial community structure through selective pressures (Li et al. 2019; Xu et al. 2019).

Notably, environment PC1, an indicator of variables influencing sedimentary bacterial communities that integrates the influence of submerged macrophyte community, is pivotal in shaping sediment bacterial communities. Higher values of environment PC1 were associated with increased sedimentary nutrient concentrations, which significantly influenced bacterial community structure. The null model analysis revealed a gradient in the influence of selection processes (both heterogeneous and homogeneous) and neutral processes (dispersal limitation and homogenising dispersal) on microbial community assembly. In our study, the increased nutrient concentrations in the shallow zone, related to higher macrophyte coverage and diversity, likely shaped microbial communities through selection processes.

We also found that sediment bacterial communities were less diverse in the SZ (Shannon diversity = 7.00). This is consistent with previous findings that eutrophication might enhance the role of selection processes in shaping microbial community assembly (Zhang, Chen, et al. 2021), leading to predictable and homogeneous community compositions (Zhou et al. 2023).

4.3 | Microbial co-Occurrence Patterns and Keystone Species

Our network analysis revealed non-random co-occurrence patterns among microbial taxa, indicating complex interactions within sediment microbial communities. The results showed that the five network modules corresponded well with the three areas. This indicates that the modular structure or property of sediment bacterial communities is sensitive to changes in nutrient levels (Li et al. 2015; Liao et al. 2021). We found that Bacteroidota and Chloroflexi exhibited higher degrees and node numbers in the community network of the SZ and MZ, respectively. Bacteroidota is typically more abundant in nutrient-rich environments, where it contributes to the breakdown of complex polysaccharides, proteins, and lipids (e.g., plant residues), releasing nutrients to support the growth of other bacteria (Pilgrim

et al. 2022; Yang et al. 2022). Chloroflexi is particularly involved in photoautotrophic activity and exhibits a lower growth rate compared to other heterotrophic bacteria (Chiriac et al. 2023; van de Meene et al. 2007). As sediment nutrient levels increase, the importance of primary producers in microbial communities declines due to competition stress from faster-growing, opportunistic taxa (Coelho et al. 2015; Sorensen and Shade 2020).

The results showed that the node number and degree of sub-networks were significantly negatively correlated with environment PC1 ($p < 0.05$), suggesting a potential loss of diversity or interaction strength within the network, likely due to increased nutrient availability or external stressors (Faust and Raes 2012; Ramoneda et al. 2023). The decomposition of submerged plant detritus releases dissolved organic carbon and other nutrients into the sediment, providing a rich source for microbial communities (Chao et al. 2021). This process enhances microbial growth and activity by supplying easily metabolizable compounds such as amino acids and simple sugars, likely supporting a higher proportion of opportunistic and fast-growing microbial taxa (He et al. 2023). These taxa often exhibit simplified interaction networks, as they rely less on mutualistic and cooperative interactions compared to nutrient-deficient microbes that depend on more complex interactions to obtain scarce resources (Ramoneda et al. 2023; Zengler and Zaramela 2018). Our results also indicate that half of the keystone species significantly decreases along environmental PC1. Keystone species play vital roles in nutrient cycling and maintaining network stability, and their decline may result in the loss of essential ecosystem services provided by microbial communities (Chen et al. 2024; Wang et al. 2023). The decline in the abundance of key species may be associated with increased nutrient availability in the community and a decrease in the need for cross-feeding (Murphy et al. 2021), highlighting the impact of rising nutrient levels on microbial community structure (Zhang et al. 2022).

4.4 | Bacterial Metabolic Functions

The functional analysis using PICRUSt2 and KEGG pathway annotation revealed significant shifts in bacterial metabolic functions along the environmental gradient represented by environment PC1. Pathways including carbohydrate metabolism, cell growth, and energy metabolism showed a significant increase with increasing nutrient levels along environment PC1. The rising carbohydrate metabolism pathways indicate that the sediment bacterial community can more efficiently use these resources to generate energy and essential metabolic intermediates (Liang et al. 2024). The enhanced cell growth pathways indicate that microbes take advantage of the nutrient-rich conditions to proliferate rapidly (Srour et al. 2020). Energy metabolism pathways are elevated as the microbial communities exploit the available nutrients to optimize their energy production processes (Srour et al. 2020; Yazhini et al. 2024). The increase in these pathways indicates that high-nutrient environments related to submerged macrophytes are associated with microbial communities that show higher resource use efficiency, faster growth, and simpler, more efficient metabolic processes. Consistent with the results of co-occurrence network analysis, the decline in their abundance will lead to a reduction in interspecific interactions (e.g., cross-feeding of amino acids

and vitamins) among microorganisms (Ramoneda et al. 2023; Zengler and Zaramela 2018).

However, we observed the pathway related to methane metabolism did not significantly increase along environment PC1, potentially due to the increase in nitrate (NO_3^- -s) levels. Elevated nitrate concentrations create more oxidizing conditions, which inhibit methanogenesis, as methanogens thrive in strictly anaerobic, low-redox environments (Chen et al. 2022; Vigderovich et al. 2023).

5 | Conclusion

Our results characterize the relationships between long-established submerged macrophyte communities and environmental parameters as well as sediment microbial communities in Dongshan Bay, Taihu Lake. This study demonstrates the association of macrophytes with key ecosystem processes, including nutrient cycling, sediment stabilization, and sediment bacterial community structure. The established lush submerged macrophyte communities, particularly in the shallow zone, were found to be correlated with improved water quality, with reduced suspended solids and nutrient concentrations, alongside increased sedimentary nutrient content. The PLS-PM analysis revealed that the submerged macrophyte community structure—including coverage, species richness, and Shannon index—was related to sediment nutrient accumulation through potential direct and indirect processes, such as the deposition of plant detritus and the modification of local hydrodynamic conditions. These changes, in turn, strengthen homogenizing selection on sediment bacterial communities, ultimately resulting in reduced microbial diversity. Network analysis suggested that macrophyte-associated conditions coincided with lower network complexity and shifts in keystone species abundance in sediment. Functional predictions indicated a potential for bacterial growth-related pathways and simpler metabolic processes in these areas, whereas greenhouse gas-related pathways showed no significant increase with macrophyte expansion. Overall, these findings highlight the complex interactions between submerged macrophytes, environmental parameters, and sediment microbial communities. These insights advance our understanding of macrophyte-dominated ecosystems and may guide long-term restoration strategies for eutrophic lakes.

Author Contributions

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** fwb70120-sup-0001-AppendixS1.docx.